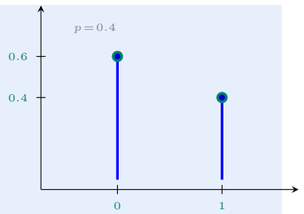
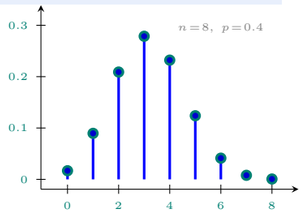
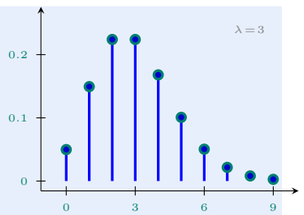
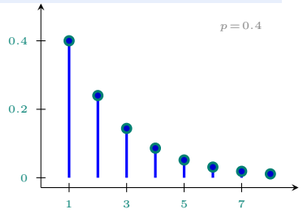
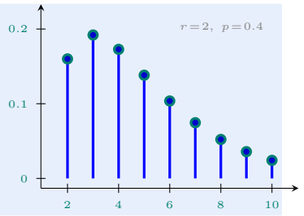
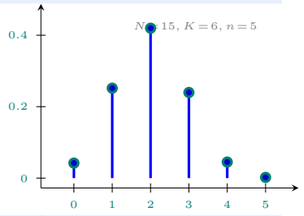
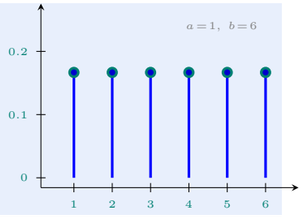
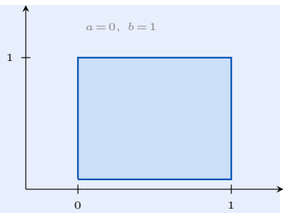
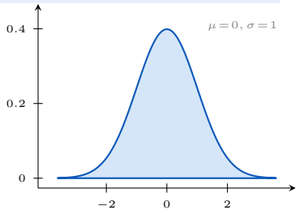
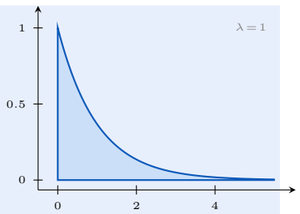
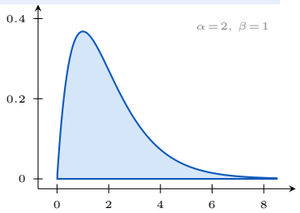
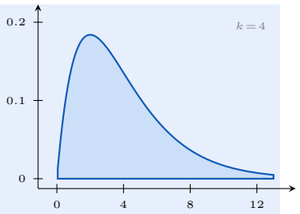
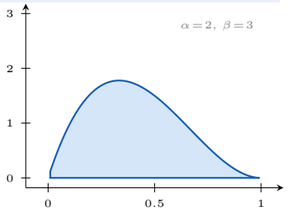
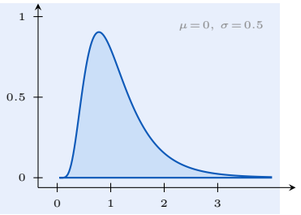
Catálogo de Distribuciones de Probabilidad

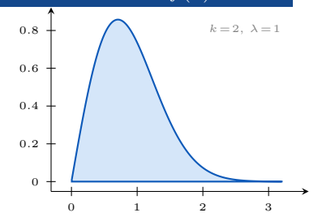
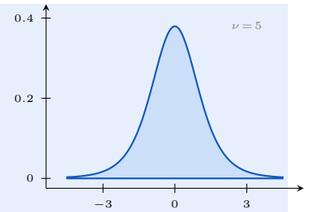
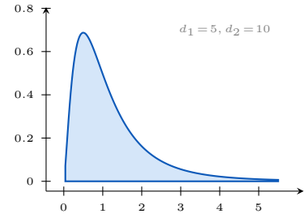
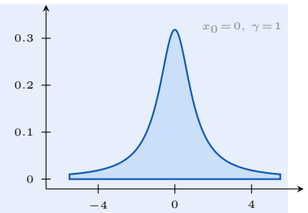
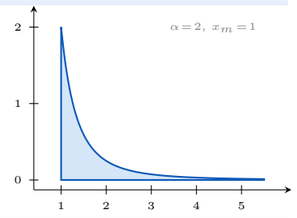
Funciones de masa/densidad, acumulada, momentos y función generadora de momentos (FGM)

I. Distribuciones Discretas

Distribución	Notación	F.M.P. $p(x)$	F.D.A. $F(x)$	Media / Varianza	FGM $M_X(t)$	Gráfica $p(x)$
Bernoulli	$X \sim \mathrm{Ber}(p)$ $p \in (0,1)$	$p^x(1-p)^{1-x}$, $x \in \{0,1\}$	$F(x) = \begin{cases} 0 & x<0 \\ 1-p & 0 \le x < 1 \\ 1 & x \ge 1 \end{cases}$	$\mu = p$ $\sigma^2 = p(1-p)$ $\gamma_1 = \frac{1-2p}{\sqrt{p(1-p)}}$	$M_X(t) = 1-p+pe^t$ $t \in \mathbb{R}$	
Binomial	$X \sim \mathrm{Bin}(n,p)$ $n \in \mathbb{N}$, $p \in (0,1)$	$\binom{n}{x} p^x (1-p)^{n-x}$, $x = 0,1,\ldots,n$	$I_{1-p}(n-\lfloor x \rfloor,\ \lfloor x \rfloor+1)$	$\mu = np$ $\sigma^2 = np(1-p)$ $\gamma_1 = \frac{1-2p}{\sqrt{np(1-p)}}$	$M_X(t) = (1-p+pe^t)^n$ $t \in \mathbb{R}$	
Poisson	$X \sim \mathrm{Poi}(\lambda)$ $\lambda > 0$	$\frac{e^{-\lambda}\lambda^x}{x!}$, $x = 0,1,2,\ldots$	$e^{-\lambda} \sum_{k=0}^{\lfloor x \rfloor} \frac{\lambda^k}{k!}$	$\mu = \lambda$ $\sigma^2 = \lambda$ $\gamma_1 = \lambda^{-1/2}$, $\gamma_2 = \lambda^{-1}$	$M_X(t) = e^{\lambda(e^t-1)}$ $t \in \mathbb{R}$	
Geométrica	$X \sim \mathrm{Geo}(p)$ $p \in (0,1)$	$(1-p)^{x-1}p$, $x = 1,2,3,\ldots$	$1-(1-p)^{\lfloor x \rfloor}$	$\mu = \frac{1}{p}$ $\sigma^2 = \frac{1-p}{p^2}$ $\gamma_1 = \frac{2-p}{\sqrt{1-p}}$	$\frac{pe^t}{1-(1-p)e^t}$, $t < -\ln(1-p)$	
Binomial Negativa	$X \sim \mathrm{BN}(r,p)$ $r \in \mathbb{N}$, $p \in (0,1)$	$\binom{x-1}{r-1} p^r (1-p)^{x-r}$, $x = r, r+1, \ldots$	$I_p(r,\ \lfloor x \rfloor - r + 1)$	$\mu = \frac{r}{p}$ $\sigma^2 = \frac{r(1-p)}{p^2}$ $\gamma_1 = \frac{2-p}{\sqrt{r(1-p)}}$	$\left(\frac{pe^t}{1-(1-p)e^t}\right)^r$, $t < -\ln(1-p)$	
Hiperge-ométrica	$X \sim$ $\mathrm{HG}(N,K,n)$	$\frac{\binom{K}{x}\binom{N-K}{n-x}}{\binom{N}{n}}$	No tiene forma cerrada simple	$\mu = n\frac{K}{N}$ $\sigma^2 = n\frac{K(N-K)(N-n)}{N^2(N-1)}$	No tiene forma cerrada simple	
Uniforme Discreta	$X \sim \mathrm{U}\{a,b\}$ $a,b \in \mathbb{Z},\ a \le b$	$\frac{1}{b-a+1}$, $x = a, a+1, \ldots, b$	$\frac{\lfloor x \rfloor - a + 1}{b-a+1}$	$\mu = \frac{a+b}{2}$ $\sigma^2 = \frac{(b-a+1)^2-1}{12}$ $\gamma_1 = 0$	$\frac{e^{at}-e^{(b+1)t}}{(b-a+1)(1-e^t)}$	

II. Distribuciones Continuas

Distribución	Notación	F.D.P. $f(x)$	F.D.A. $F(x)$	Media / Varianza	FGM $M_X(t)$	Gráfica $f(x)$
Uniforme Continua	$X \sim \mathrm{U}(a,b)$ $a < b$	$\frac{1}{b-a}, \quad x \in [a,b]$	$\frac{x-a}{b-a}$	$\mu = \frac{a+b}{2}$ $\sigma^2 = \frac{(b-a)^2}{12}$ $\gamma_1 = 0,\ \gamma_2 = -\frac{6}{5}$	$\frac{e^{tb}-e^{ta}}{t(b-a)}, \quad t \neq 0$ $M_X(0) = 1$	
Normal	$X \sim \mathcal{N}(\mu, \sigma^2)$ $\mu \in \mathbb{R},\ \sigma > 0$	$\frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$	$\Phi\left(\frac{x-\mu}{\sigma}\right) = \frac{1}{2}\left[1 + \mathrm{erf}\left(\frac{x-\mu}{\sigma\sqrt{2}}\right)\right]$	$\mu = \mu$ $\sigma^2 = \sigma^2$ $\gamma_1 = 0,\ \gamma_2 = 0$	$M_X(t) = e^{\mu t + \frac{1}{2}\sigma^2 t^2}$ $t \in \mathbb{R}$ $E[X^{2k}] = (2k-1)!!\,\sigma^{2k}$	
Exponencial	$X \sim \mathrm{Exp}(\lambda)$ $\lambda > 0$	$\lambda e^{-\lambda x}, \quad x \geq 0$	$1 - e^{-\lambda x}$	$\mu = \frac{1}{\lambda}$ $\sigma^2 = \frac{1}{\lambda^2}$ $\gamma_1 = 2,\ \gamma_2 = 6$	$M_X(t) = \frac{\lambda}{\lambda - t}, \quad t < \lambda$ $E[X^n] = \frac{n!}{\lambda^n}$	
Gamma	$X \sim \Gamma(\alpha, \beta)$ $\alpha, \beta > 0$	$\frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}, \quad x > 0$	$\frac{\gamma(\alpha,\ \beta x)}{\Gamma(\alpha)}$	$\mu = \frac{\alpha}{\beta}$ $\sigma^2 = \frac{\alpha}{\beta^2}$ $\gamma_1 = \frac{2}{\sqrt{\alpha}},\ \gamma_2 = \frac{6}{\alpha}$	$M_X(t) = \left(\frac{\beta}{\beta - t}\right)^\alpha, t < \beta$ $E[X^n] = \frac{\Gamma(\alpha+n)}{\Gamma(\alpha)\,\beta^n}$	
Chi-cuadrada	$X \sim \chi^2(k)$ $k \in \mathbb{N}$	$\frac{x^{k/2-1} e^{-x/2}}{2^{k/2}\Gamma(k/2)}, \quad x > 0$	$\frac{\gamma\left(\frac{k}{2}, \frac{x}{2}\right)}{\Gamma\left(\frac{k}{2}\right)}$	$\mu = k$ $\sigma^2 = 2k$ $\gamma_1 = \sqrt{\frac{8}{k}},\ \gamma_2 = \frac{12}{k}$	$M_X(t) = (1-2t)^{-k/2}, t < \frac{1}{2}$ Caso especial: $\Gamma\left(\frac{k}{2}, \frac{1}{2}\right)$	
Beta	$X \sim$ $\mathrm{Beta}(\alpha, \beta)$ $\alpha, \beta > 0$	$\frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha, \beta)}, \quad x \in (0,1)$	$I_x(\alpha, \beta)$ (Beta incompleta regularizada)	$\mu = \frac{\alpha}{\alpha+\beta}$ $\sigma^2 = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$ $E[X^n] = \frac{B(\alpha+n,\ \beta)}{B(\alpha, \beta)}$	$1 + \sum_{k=1}^{\infty}\left(\prod_{r=0}^{k-1} \frac{\alpha+r}{\alpha+\beta+r}\right)\frac{t^k}{k!}$	
Log-Normal	$X \sim \mathrm{LN}(\mu, \sigma^2)$ $\mu \in \mathbb{R},\ \sigma > 0$	$\frac{1}{x\sigma\sqrt{2\pi}} \exp\left(-\frac{(\ln x - \mu)^2}{2\sigma^2}\right), x > 0$	$\Phi\left(\frac{\ln x - \mu}{\sigma}\right)$	$E[X] = e^{\mu + \sigma^2/2}$ $\mathrm{Var} = (e^{\sigma^2} - 1)e^{2\mu+\sigma^2}$ $E[X^n] = e^{n\mu + n^2\sigma^2/2}$	No existe para $t > 0$ (los momentos existen para todo n)	

Distribución	Notación	F.D.P. $f(x)$	F.D.A. $F(x)$	Media / Varianza	FGM $M_X(t)$	Gráfica $f(x)$
Weibull	$X \sim \mathrm{W}(k,\lambda)$ $k,\lambda > 0$	$\frac{k}{\lambda}\left(\frac{x}{\lambda}\right)^{k-1} e^{-(x/\lambda)^k}, x \geq 0$	$1 - e^{-(x/\lambda)^k}$	$\mu = \lambda\,\Gamma\left(1+\frac{1}{k}\right)$ $\sigma^2 = \lambda^2\left[\Gamma\left(1+\frac{2}{k}\right) - \Gamma^2\left(1+\frac{1}{k}\right)\right]$ $E[X^n] = \lambda^n\,\Gamma\left(1+\frac{n}{k}\right)$	$\sum_{n=0}^{\infty} \frac{t^n \lambda^n}{n!}\Gamma\left(1+\frac{n}{k}\right)$	
***t* de Student**	$X \sim t(\nu)$ $\nu > 0$	$\frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\nu\pi}\,\Gamma\left(\frac{\nu}{2}\right)}\left(1+\frac{x^2}{\nu}\right)^{-\frac{\nu+1}{2}}$	$\frac{1}{2} + x\frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\nu\pi}\,\Gamma\left(\frac{\nu}{2}\right)}\,{}_2F_1\left(\frac{1}{2}, \frac{\nu+1}{2}; \frac{3}{2}; -\frac{x^2}{\nu}\right)$	$\mu = 0\ (\nu > 1)$ $\sigma^2 = \frac{\nu}{\nu-2}\ (\nu > 2)$ $\gamma_1 = 0,\ \gamma_2 = \frac{6}{\nu-4}\ (\nu > 4)$	No existe en forma cerrada	
***F* de Snedecor**	$X \sim F(d_1,d_2)$ $d_1,d_2 > 0$	$\frac{\sqrt{\frac{(d_1x)^{d_1}d_2^{d_2}}{(d_1x+d_2)^{d_1+d_2}}}}{x\,B\left(\frac{d_1}{2},\frac{d_2}{2}\right)}, x > 0$	$I_{\frac{d_1x}{d_1x+d_2}}\left(\frac{d_1}{2},\frac{d_2}{2}\right)$	$\mu = \frac{d_2}{d_2-2}\ (d_2 > 2)$ $\sigma^2 = \frac{2d_2^2(d_1+d_2-2)}{d_1(d_2-2)^2(d_2-4)}\ (d_2 > 4)$	No existe (Si $Y_i \sim \chi^2(d_i)$, entonces $\frac{Y_1/d_1}{Y_2/d_2} \sim F(d_1,d_2)$)	
Cauchy	$X \sim \mathrm{Ca}(x_0,\gamma)$ $x_0 \in \mathbb{R},\ \gamma > 0$	$\frac{1}{\pi\gamma\left[1+\left(\frac{x-x_0}{\gamma}\right)^2\right]}$	$\frac{1}{\pi}\arctan\left(\frac{x-x_0}{\gamma}\right) + \frac{1}{2}$	μ no existe σ^2 no existe La mediana es x_0	No existe (Las colas pesadas impiden la convergencia de $E[e^{tX}]$)	
Pareto	$X \sim \mathrm{Pa}(\alpha,x_m)$ $\alpha,x_m > 0$	$\frac{\alpha\,x_m^\alpha}{x^{\alpha+1}},\quad x \geq x_m$	$1 - \left(\frac{x_m}{x}\right)^\alpha$	$\mu = \frac{\alpha x_m}{\alpha-1}\ (\alpha > 1)$ $\sigma^2 = \frac{x_m^2\alpha}{(\alpha-1)^2(\alpha-2)}\ (\alpha > 2)$ $E[X^n] = \frac{\alpha x_m^n}{\alpha-n}\ (n < \alpha)$	No existe para $t > 0$ (Distribución de cola pesada; codifica la "ley de Pareto 80/20")	

Notación: $\Gamma(\cdot)$: función Gamma; $B(a,b) = \Gamma(a)\Gamma(b)/\Gamma(a+b)$: función Beta; $I_x(a,b)$: función Beta incompleta regularizada; $\gamma(\cdot,\cdot)$: función Gamma incompleta inferior; $\Phi(\cdot)$: f.d.a. de la normal estándar; γ_1: asimetría; γ_2: curtosis excesiva.